Processes of a Project

Introduction

- Project Management is the application of knowledge, skills, tools and techniques to project activities to meet project objectives
- It is accomplished through processes using knowledge, skills, tools and techniques with the received inputs and generate outputs
- A process is a set of interrelated actions and activities that are performed to achieve a pre-specified set of products, results or services

Processes contd..

- The project processes are performed by the project team and can be categorized at high level into two types
- Processes associated with each other by their performance for an integrated purpose; this type are common to most projects
- Product oriented processes specify and create the project's product; they vary depending on the application
- Both of the above type of processes overlap and interact; for example to define the scope of the project, understanding of how to create the product is required

Process groups

- Processes can be divided into different groups in terms of the integration between the processes, the interaction within them and the purposes they serve
- Initiating process groups
- Planning process groups
- Executing process groups
- Monitoring and controlling process groups
- Closing process groups

Initiating process groups

- This group consists of processes that facilitate the formal authorization to start a new project or a project phase
- The group includes the following processes

1. **Develop Process Charter**

   Initial assumptions, decisions made and constraints are documented in process charter

   The project charter links the project to the ongoing work of the organization and authorizes the project

Initiating process groups contd..

2. **Develop Preliminary Project Scope**

- This process is responsible for producing preliminary high level definition of the project
- Documents are made for the following

- project and deliverable requirements
- product requirements
- boundaries of the project
- methods of acceptance
- high level scope control

Planning Process Group

- Includes the following Project Management processes

1. Develop Project Management Plan

This process is responsible for planning, executing, monitoring, controlling and closing of the project

2. **Scope planning**

This process is needed for creating a project scope management plan that includes

- scope definition, verification, control
- work breakdown structure creation and definition

3. **Scope Definition**

Necessary for developing a detailed project scope statement which is the basis for future scope decision

Planning Process Group contd..

4. **Create work breakdown structure**

It is used for subdividing the major project deliverables, and project work into smaller and more manageable components

5. **Activity Definition**

Identify the specific actions that need to be performed to produce the various project deliverables

6. **Activity Sequencing**

Identifying and documenting dependencies among scheduled activities

Planning Process Group contd..

7. Activity Resource Estimation

Estimating the type and quantities of resources required to perform each scheduled activity

8. **Activity Duration Estimation**

Estimating the number of work periods that will be needed to complete individual scheduled activities

9. **Schedule Development**

Analyzing activity sequence, duration, resource requirement and schedule constraints to create the project schedule

Planning Process Group contd..

10. Cost Estimating

Developing an approximation of the costs of the resources needed to complete project activities

11. Cost Budgeting

Aggregating the estimated costs of the individual activities or work packages to establish a cost baseline

12. Quality Planning

Indentifying which quality standards are relevant to the project and determining how to satisfy them

Planning Process Group contd..

13. Human Resource Planning

Identifying and documenting project roles, responsibilities, and reporting relationships and also creating the staffing management plan

14. Communications Planning

Determining the information and communication needs of the project stakeholders

15. Risk Management Planning

Deciding how to approach, plan, and execute the risk management activities for a project

Planning Process Group contd..

16. Risk Identification

Determining which risks might affect the project and documenting their characteristics

17. Qualitative Risk Analysis

For prioritizing risks for subsequent analysis or action by assessing and combining the probability of occurrence and impact

18. Quantitative Risk Analysis

For numerically analyzing the effect on overall project objectives of identified risks

Planning Process Group contd..

19. Risk Response Planning

For developing options and actions to enhance opportunities and to reduce threats to project objectives

20. Plan Purchases and Acquisitions

For determining what to purchase, or acquire and determining when and how

21. Plan Contracting

For documenting products, services and results requirements and identifying potential sellers

3. Executing Process Group

- Consists of processes used to complete the work defined in the project management plan to accomplish the project objectives
- The processes in this group are as follows

1. Direct and Manage Project Execution

   Directing the various technical and organizational interfaces that exist in the project to execute the work defined in the project management plan

Executing Process Group contd..

2. Perform Quality Assurance

Applying the planned, systematic quality activities to ensure that the project employs all processes needed to meet requirements

3. Acquire Project Team

Obtaining the human resources needed to complete the project

4. Develop Project Team

Improving the competencies and the interactions of the team members to enhance project performance

Executing Process Group contd..

5. Information Distribution

   Making information available to project stakeholders in a timely manner

6. Request Seller Responses

   Obtaining information, quotations, bids, offers or proposals

7. Select sellers

   Reviewing offers, choosing from among potential sellers and negotiating a written contract with the seller

4. Monitoring and Controlling Process Group

- It consists of the processes that are performed to observe project execution to identify potential problems in time so that corrective action can be taken to control the execution of the project
- It consists of the following processes

1. Monitor and control project work

Collecting, measuring and disseminating performance information, assessing progress, and risk monitoring

Performance reports provide information on the project with respect to scope, schedule, cost, resource, quality and risk

Monitoring and Controlling Process Group contd..

2. Integrated Change Control

Controlling factors that create changes to make sure those changes are beneficial and managing approved changes

This process is performed throughout the project from initiation to closure

3. Scope Verification

This process formalizes acceptance of the completed project deliverables

Monitoring and Controlling Process Group contd..

4. Scope Control

This process is necessary for controlling changes to the project scope

5. Schedule Control

Controlling changes to project schedule

6. Cost Control

Influencing the factors that create variances and controlling changes to the project budget

Monitoring and Controlling Process Group contd..

7. Perform Quality Control

Monitoring specific project result to determine whether they comply with relevant quality standards and identifying ways to eliminate causes of unsatisfactory performance

8. Manage Project Team

Tracking team member performance providing feedback, resolving issues, and coordinating changes to enhance project performance

Monitoring and Controlling Process Group contd..

9. Performance Reporting

This process is necessary for collecting and distributing performance information

It includes status reporting, progress measurement and forecasting

10. Manage Stakeholders

Manage communications to satisfy the requirements of and resolve issues with project stakeholders

Monitoring and Controlling Process Group contd..

11. Risk Monitoring and Control

Tracking identified risks, monitoring residual risks, identifying new risks, executing risk response plan, and evaluating their effectiveness throughout the project life cycle

12. Contract Administration

This process is necessary for managing the contract and relationship between the buyer and seller

Reviewing and documenting how a seller is performing or performed

5. Closing Process Group

- This group includes the processes needed to formally terminate all activities of a project or a project phase
- It includes the following processes

1. Close Project

This is necessary to finalize all activities across all of the process groups to formally close a project or a project phase

2. Contract Closure

This is necessary for completing and settling each contract

- All of the above processes may not be needed for all projects

Process Interactions

- The Process Groups are linked by the objectives they produce
- The output of one process may be the input of another process or the output is a deliverable of the project
- The Planning process group provides the executing process group a document project management plan and project scope statement
- Also updates the project management plan as the project progresses

Process Interactions contd..

- The process groups are rarely discrete or one time events
- They are overlapping activities that occur at varying levels of intensity throughout the project
- If the project is divided into phases, the process groups interact within a project phase and also may across the project phases

Project Milestones

- All projects will need to have some fixed milestones or time boxes if progress towards a planned delivery date is to be maintained
- The project customer or project sponsor often decides the key events or major milestones for the completion of certain deliverables by a specified date
- The project manager decides the minor milestones or the key events for the routine monitoring activities

Checkpoint Activities

- Sequencing activities for a project involves starting a new activity as soon as the preceding ones upon which it depends have been completed
- Sometimes instead of sequence of activities, the project is divided into stages and checkpoint activities are introduced
- These checkpoint activities which are the key activities represent the completion of important stages of the project
- The check point activities are often used as useful milestones

Major Milestones, Minor Milestones, Periodic Status Assessments

- The four Major Milestones occur at the transition points between life cycle phases
- Inception, Elaboration, Construction, Transition
- The four types of major milestones are

1. Life cycle objectives Milestone
2. Life cycle Architecture Milestone
3. Initial Operational Capacity Milestone
4. Product Release Milestone

Life cycle objectives Milestone

- The goal is to present to all stakeholders a recommendation on how to proceed with development including a plan, estimated cost and schedule and expected benefits
- A successfully completed life cycle objectives milestone will result in authorization from all stakeholders

Life cycle Architecture Milestone

- The primary goal is to demonstrate an executable architecture to all stakeholders
- A more detailed plan for construction phase is presented for approval
- A successfully completed Life cycle Architecture Milestone will result in authorization from stakeholders to proceed with the construction phase

Life cycle Architecture Milestone contd..

- The content of the milestone should include the following
- An overview of the current present state
- A configuration controlled set of information available electronically or in hard copy
- An executable demonstration of capability

Initial Operational Capability Milestone

- The goals are to assess the readiness of the software to begin the transition into customer/user sites and to authorize the start of acceptance testing
- Issues are addressed concerning installation instructions, software version descriptions user and operator manuals etc.
- Software quality metrics are reviewed to determine whether quality is sufficient for transition

Product Release Milestone

- The Product Release Milestone occurs at the end of the transition phase
- The goal is to assess the completion of the software and its transition to the support organization
- The results of acceptance testing are reviewed and all open issues are addressed

Minor Milestones

- The number of informal milestones needed depends on the content and length of the iteration or tasks
- For most iterations which have one-month to six-month duration, only two minor milestones are needed: the iteration readiness review and the iteration assessment review

Iteration Readiness Review

- This informal milestone is conducted at the start of each iteration to review the detailed plan and the evaluation criterion that have been allocated
- Iteration Assessment Review: This informal milestone is conducted at the end of each iteration to assess the degree to which the iteration achieved its objectives and satisfied its evaluation criteria

Periodic Status Assessments

- Managing risks requires continuous attention to all the interacting activities of a software development effort
- Periodic Status Assessments are management reviews conducted at regular intervals (monthly, quarterly) to address progress and quality indicators, ensure continuous attention to project dynamics, and maintain open communications among all stakeholders
- The important objective of these assessments is to ensure that the expectations of all stakeholders are synchronized and consistent

Periodic Status Assessments contd..

- Status assessments provide the following
- A mechanism for openly addressing, communicating, and resolving management issues, technical issues, and project risks
- Objective data derived directly from ongoing activities and evolving product configurations
- A mechanism for disseminating process, progress, quality trends, practices, and experience information to and from all stakeholders in an open forum

Additional Reference

- Walker Royce, Software Project Management A Unified Framework, Pearson Education